

Abhinav Jha

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github.com/Abhinav180305

Education

- **VIT Bhopal University**
B.Tech in Computer Science & Engineering; CGPA: 8.54/10.0 2023–2027

Technical Skills

- **Programming Languages:** C++, Python
- **AI / Machine Learning:** Machine Learning, Deep Learning, Computer Vision, NLP, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, NumPy
- **Generative AI:** LLMs, RAG, LangChain, LangGraph, LangSmith, Prompt Engineering, LLM APIs
- **Backend / Web:** FastAPI, HTML, CSS, JavaScript
- **Tools & Platforms:** Streamlit, Google Colab, Matplotlib

Project Experience

Driver Drowsiness Detection System (ML + Computer Vision) github.com/Abhinav180305/Driver-Safety-System

- Developed a real-time driver monitoring system using computer vision and deep learning (YOLO) to detect eye closure and fatigue patterns.
- Implemented OpenCV-based video processing and optimized model inference for real-time performance, ensuring accurate detection under varying lighting and driving conditions.
- Created a custom dataset for the project and built an instant-alert feature to reduce accident risk.
- Trained and evaluated the detection model using TensorFlow and Keras, tuning hyperparameters to improve accuracy across diverse test scenarios.

AI Data Analyst github.com/Abhinav180305/Ai-Data-Analyst

- Developed an AI-powered data analyst that accepts CSV files and automates analysis, visualization, and prediction.
- Built a benchmarking pipeline to train and compare 4–5 ML models using relevant performance metrics.
- Generated feature-level plots, a general dashboard, and a correlation heatmap to highlight key patterns.
- Added natural-language queries that convert dataset questions into executable Python or SQL code.

YouTube Video Q&A — Chrome Extension + Local Backend (RAG) github.com/Abhinav180305/YouTube-AI-Study-Assistant

- Built a Chrome extension that injects a floating Q&A panel onto YouTube watch pages, letting users ask questions about the video they are currently watching and get answers generated from its full transcript.
- Built a FastAPI backend implementing a retrieval-augmented generation (RAG) pipeline: fetched transcripts (youtube-transcript-api, with a yt-dlp fallback), chunked and embedded the text, and answered queries using a Groq-hosted LLM.
- Designed the system so the extension never calls an AI API directly — all transcript retrieval, embedding, and LLM calls run through the local backend service, keeping the architecture modular and API keys secure.

Certifications

- Applied Machine Learning in Python — University of Michigan
- Oracle Cloud Infrastructure Certified AI Foundations Associate
- TCS iON Career Edge - AI Foundation